

How chemical properties of red wine affects its alcohol content

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Motivation

- To predict the total alcohol content base off of existing chemical compounds, such as total ph, total sulphur dioxide, and density of the must(Fermenting Juice).
- By learning the relationship between these features with the production of alcohol ABV. We can more accurately predict the total ABV more easily.
- Measuring properties such as density, ph, sulfates, etc can be faster and cheaper than legal grade assays such as distillations.

Data Source

- The dataset was obtained from Kaggle, which was sourced from the UCI machine learning, published with the CC BY 4.0 license on the UCI ML repository.
- The data was shared with the intent for education and learning
- Each datapoint contains only lab measurements of wine, with clear metadata with no sensitive information with no data inconsistency.

Project Flowchart

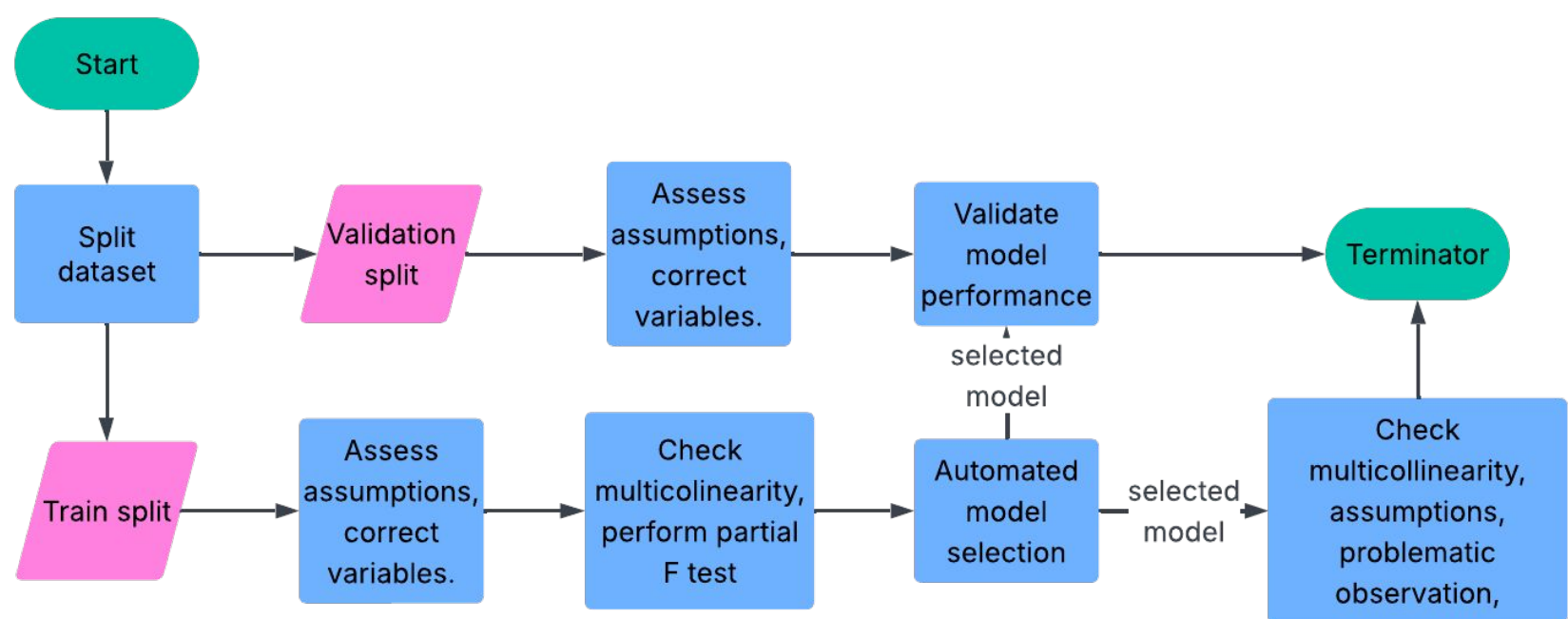


Figure 1 Project Flowchart

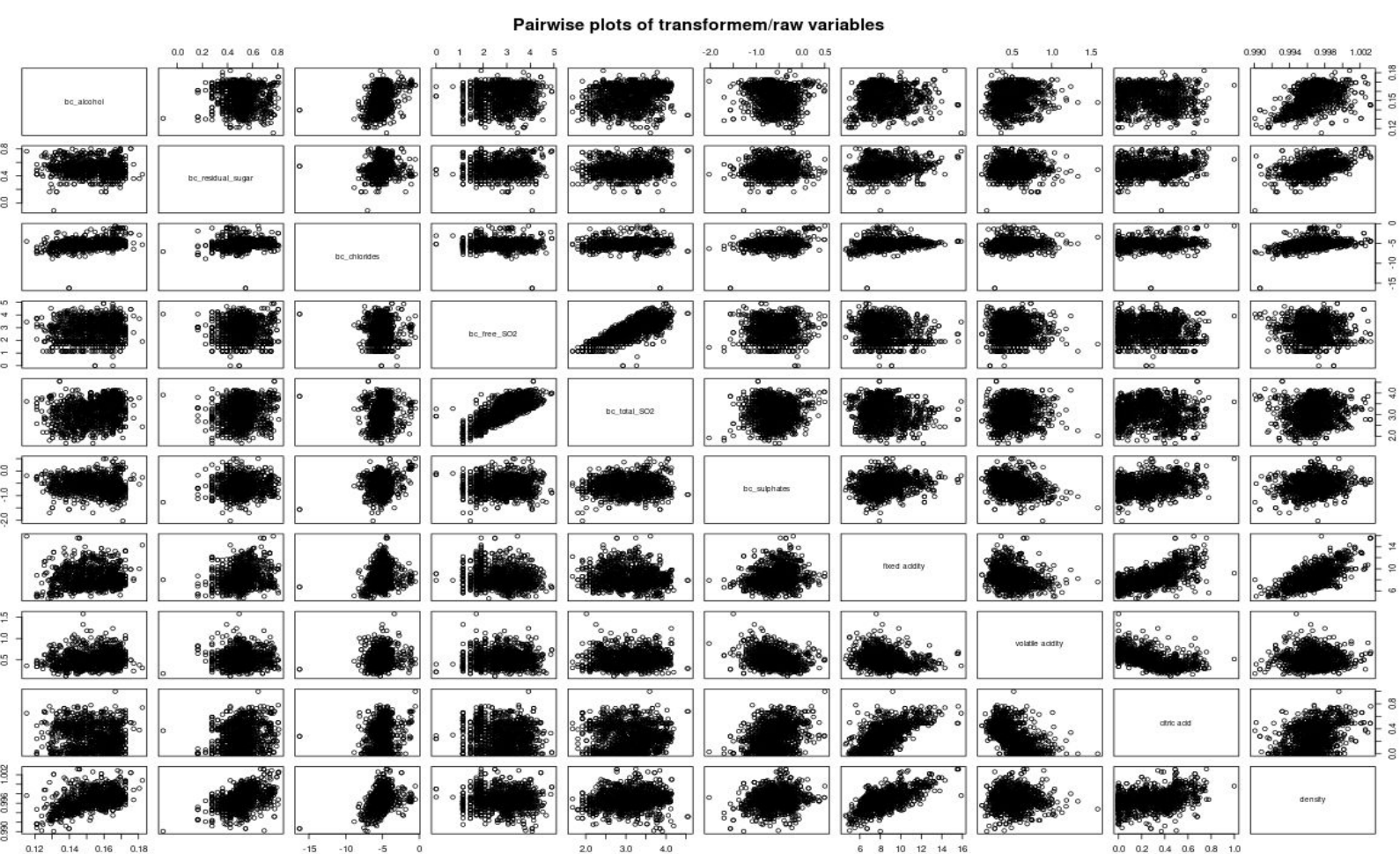


Figure 2 Transformed Predictor Pairwise Plot

Models

- Model 1: Fit preliminary model of alcohol on all predictors with interaction quality * volatile acidity
- Model 2: Transformed response alcohol with -1.9 lambda, keeping predictors constant to address right skew.
- Model 3: Added residual_sugars * chlorides interaction to model 2
- Model 4: Fit model 2 after applying box cox transformation to key predictors residual sugar, chlorides, free sulfur dioxide, and total sulfur dioxide
- Model 5: Drops bc_chlorides and bc_total_SO2 from Model 4.
- Model 6: Applied best subset selection to Model 4 using regsubsets and BIC
- Model 7: Applied stepwise selection to Model 4.

Analysis

- Partial F test on Model 2 and 3 reveals that the interaction term residual_sugars * chlorides is not statistically significant. Model 2 also has a lower BIC.
- Partial F test on Model 4 and 5 reveals that at least one of bc_chlorides and bc_total_SO2 is statistically significant.
- Running subset selection based on Model 4 using regsubsets and BIC identifies Model 6 as a low-BIC candidate that keeps quality_cat and volatile acidity:quality_cat while dropping weaker sulphur/chloride terms.
- Running stepwise selection form Model 4 achieves model with almost identical R², AIC, BIC and RMSE
- Since Model 7 contained an extra predictor than Model 6, we chose Model 6 as it is simpler with the same predictive power.

```
pH -0.0382852 0.0016415 -23.323 < 2e-16 ***
quality_catgood 0.0022547 0.0015482 1.456 0.14554
bc_residual_sugar -0.0457768 0.0019035 -24.048 < 2e-16 ***
bc_total_SO2 0.0018053 0.0003621 4.986 7.03e-07 ***
bc_sulphates -0.0068149 0.0006057 -11.251 < 2e-16 ***
quality_catbad:'volatile acidity' -0.0015997 0.0013331 -1.200 0.23036
quality_catgood:'volatile acidity' -0.0131600 0.0032973 -3.991 6.95e-05 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.006368 on 1268 degrees of freedom
Multiple R-squared: 0.7232, Adjusted R-squared: 0.721
F-statistic: 331.3 on 10 and 1268 DF, p-value: < 2.2e-16
```

Figure 3 Model 6 regression summary

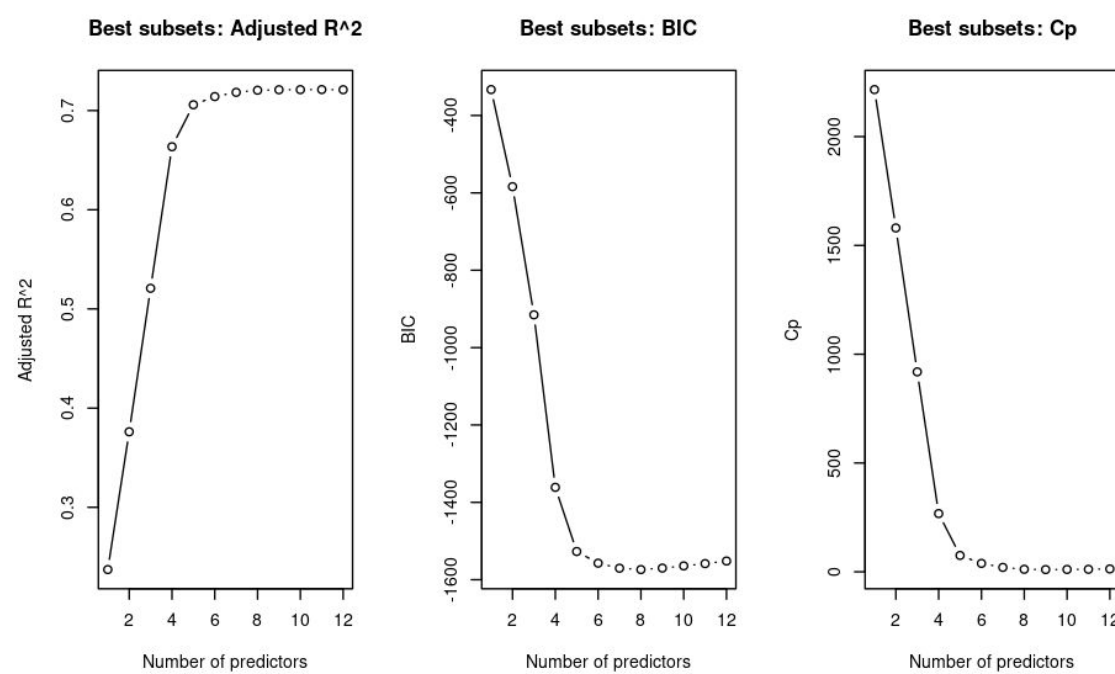


Figure 4 Model 6 Regression Subset Plot

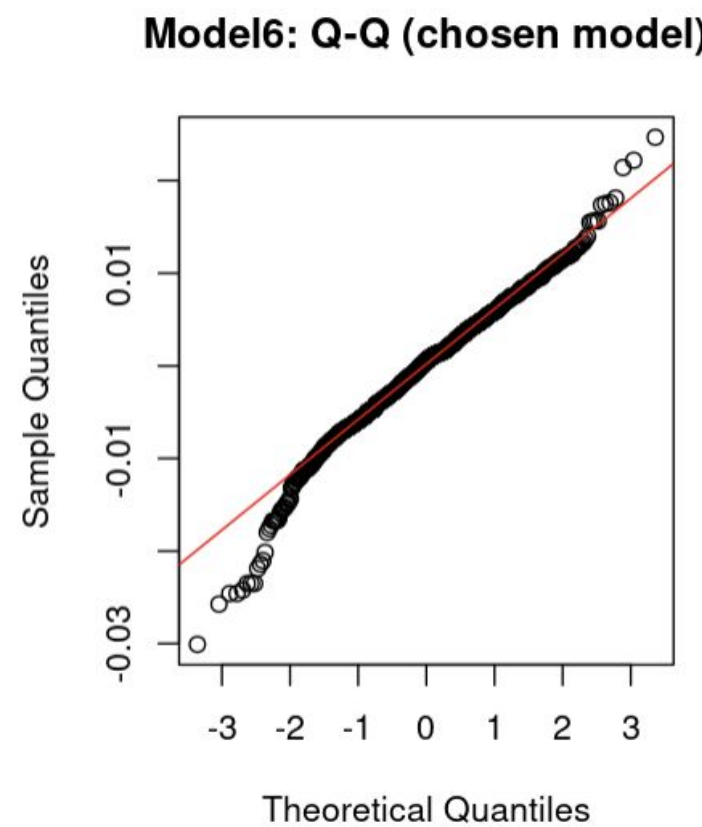


Figure 5 Model 6 QQ Plot

Conclusion

- Our transformed response term via box cox is bc_alcohol = alcohol^{-0.8}, which is a decreasing non linear transformation. This means that a negative coefficient leads to a higher mean alcohol, and conversely a positive coefficient leads to lower mean alcohol.
- The coefficient with the largest magnitude is bc_residual_sugar at around -0.046. This suggests that holding other predictors constant, wines with higher residual sugar tend to have high mean alcohol content.
- Both pH and bc_sulphates have a negative coefficient in the model of around -0.038 and -0.007, suggesting a positive relationship with alcohol content.
- bc_total_SO2 had a positive coefficient of 0.0018, suggesting higher total sulphates is slightly negative associated with alcohol content.
- For quality_catgood, a small coefficient of 0.0023 alongside with a high p value of 0.15 suggests there is a weak relationship between quality_catgood and alcohol content by itself.
- For bad wines, the interaction term quality_catbad × volatile_acidity is small and not statistically significant, so after adjusting for other variables we do not see strong evidence that volatile acidity is related to mean alcohol content in the bad group.
- For good wines, the interaction term quality_catgood × volatile_acidity has an estimated coefficient of about -0.013 on the bc_alcohol scale, meaning that higher volatile acidity is associated with higher mean alcohol content for good wines, holding the other predictors fixed

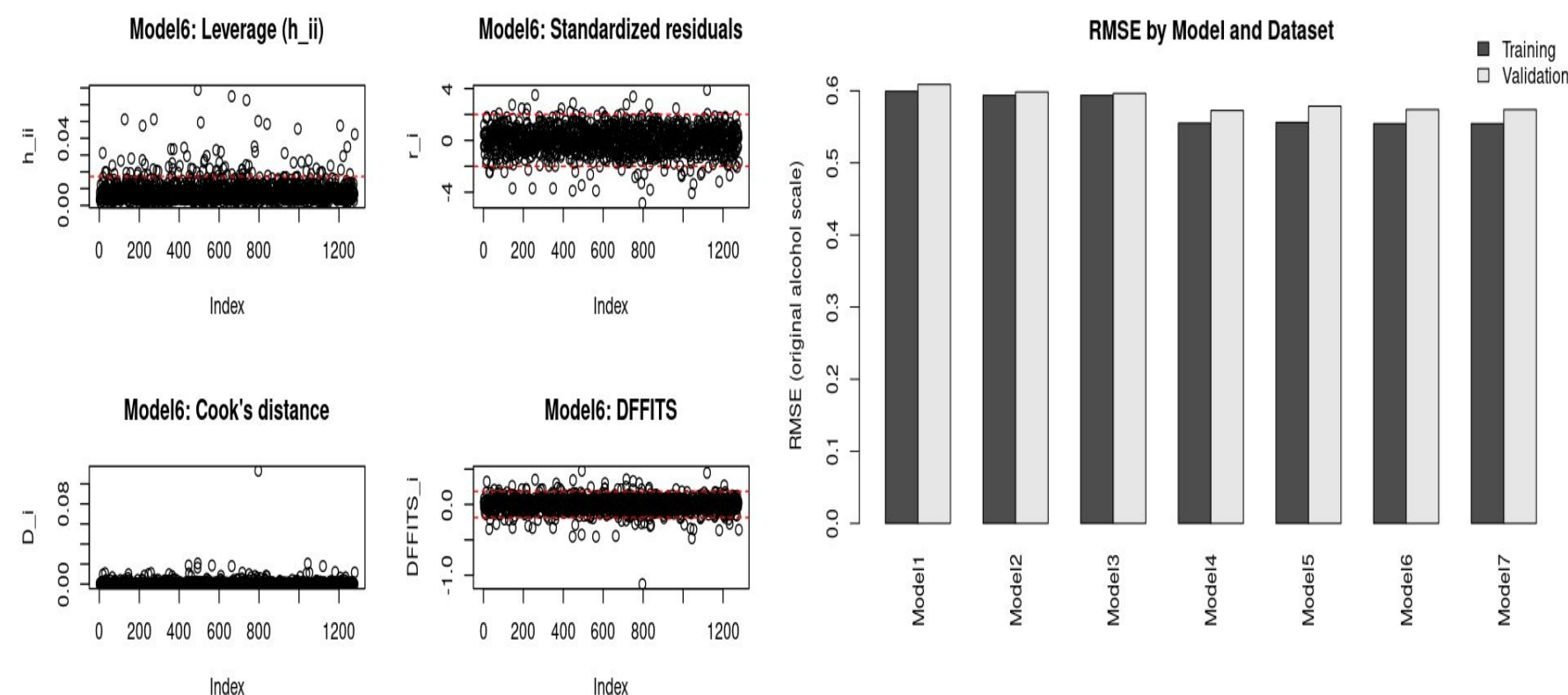


Figure 6 Model 6 influence diagnostics

Limitations

- Limited sample size of 1599
- The response is modeled on a Box–Cox scale, which makes direct interpretation on the raw alcohol scale harder
- Other properties not featured in this dataset may have relationships with alcohol content.
- The “quality” variable is based on sensory data, so it is subjective in nature.
- Our encoding of quality collapses magnitude, transforming it into a binary value. This can hide nuanced relationships between perceived quality and alcohol content.
- Several coefficients in model 6 such as quality_catgood and quality_catbad:volatile acidity are not statistically significant,